2025 International Conference on Computing, Intelligence, and Application (CIACON) | 979-8-3315-4411-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/CIACON65473.2025.11189541

AI-Driven Intelligent System for Personalized Job Recommendations Using Real-Time Skill Identification and Industry Trend Analysis

Apurba Nandi
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
apurba.nandi@uem.edu.in

Avik Kumar Das
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
avikkumar.das@uem.edu.in

Hritajit Sur
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
hritajitsur@gmail.com

Sankalpa Dutta
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
sankalpadutta04@gmail.com

Arijeet Das
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
arijeetdas999@gmail.com

Shuvam Podder
Dept. of CSE (IoT, CS & BT)
Institute of Engineering and Management, University of Engineering and Management
Kolkata, India
shuvam2424@gmail.com

***Abstract*—Navigating today's job market is challenging, especially for fresh graduates and professionals seeking the right opportunities. Traditional keyword-based job search methods often result in mismatches, making it difficult for candidates to find suitable roles and for employers to connect with the right talent. To address this, we propose an intelligent job recommendation system that leverages structured and unstructured data from job portals, recruitment databases, and candidate inquiries to provide highly relevant matches. Our approach integrates Named Entity Recognition (NER) with fuzzy matching to standardize skills, ensuring better alignment between candidate profiles and industry demands. By incorporating Levenshtein Distance for spelling correction and frequency-based skill analysis, we refine skill extraction for more precise recommendations. Additionally, Sentence-BERT (SBERT) captures semantic similarities between job descriptions and candidate qualifications, while a multi-armed bandit (MAB) algorithm optimizes job ranking. To enhance accessibility, MarianMT enables multilingual support. Experimental results show significant improvements in job matching accuracy, offering an efficient, inclusive, and scalable solution for modern recruitment challenges.**

***Index Terms*—Job Recommendation, NLP, SBERT, MarianMT, MAB, Fuzzy matching, Levenshtein Distance.**

I. Introduction

In today's fast-changing job market, finding the right opportunities has become a major challenge for job seekers, especially fresh graduates [1]. Traditional keyword-based job searches often lead to mismatches due to vague job descriptions and unclear skill requirements. As of February 2025, Europe's labor market has slowed, adding 151,000 new jobs after a revised gain of 125,000 in January. This uncertainty has left 79% of job seekers feeling anxious, with Generation Z particularly frustrated. One in five individuals who started job hunting last year are still searching, and 26% find the market especially tough. A LinkedIn report shows that 75% of professionals apply to roles where they are either underqualified or overqualified, leading to hiring inefficiencies. In India, only 42.6% of graduates are considered employable, highlighting a critical skill gap [2]. To address this, intelligent job recommendation systems must accurately analyze and standardize job skills, ensuring better alignment between candidates and industry needs.

Over the years, researchers have explored various techniques to improve job recommendations and skill extraction, ensuring a better match between job seekers and employers. Early approaches relied heavily on rule-based systems and keyword-matching techniques to identify skills from job postings (Zhang et al., 2018) [3]. While these methods worked to some extent, they lacked flexibility and often misinterpreted skills due to variations in phrasing and context. For example, the same skill could be described differently across job descriptions, leading to inconsistencies in extraction. To address these limitations, researchers introduced hybrid machine learning models that combined rule-based techniques with statistical learning for more accurate skill identification (Yu et al., 2005) [4]. These models improved precision but still struggled with domain-specific terms and inconsistencies in skill representation. With the rise of deep learning, more advanced approaches have emerged, leveraging models like Named

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.

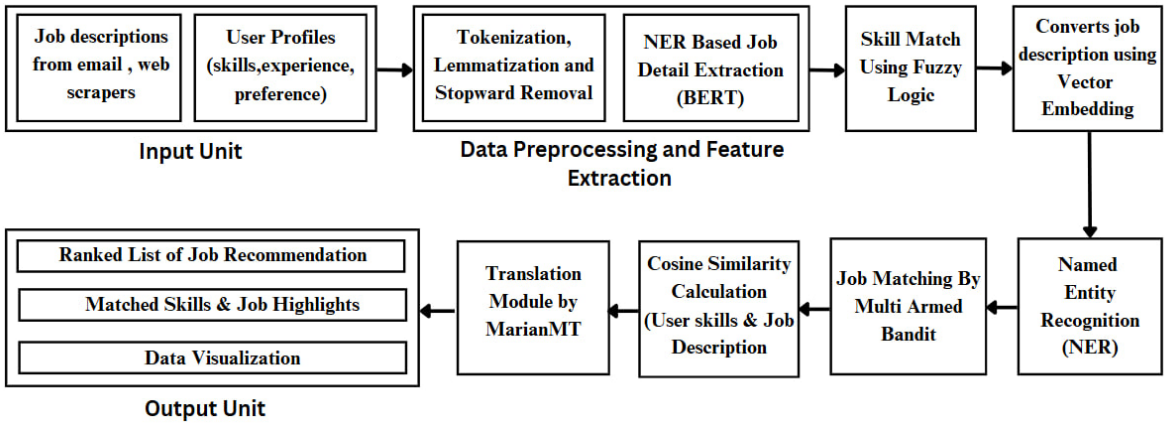


Fig. 1. Block Diagram of Proposed System

Entity Recognition (NER) using YOLOv5 and DistilBERT to extract job-related skills with higher accuracy (Kinger et al., 2024) [5]. These deep learning techniques significantly outperformed traditional methods, providing a more contextual understanding of job descriptions. However, they still faced challenges in processing large-scale job datasets and handling the evolving nature of job roles. Another promising direction in this field is the use of clustering methods to match professionals' competencies with market demands (Kobets et al., 2024) [6]. By analyzing patterns in job postings, these methods help identify trending skills and emerging career paths. Furthermore, knowledge graph-based techniques have gained attention for their ability to structure relationships between jobs and skills, enhancing recommendation accuracy (Kavas et al., 2025) [7]. AI-driven knowledge graphs further refine this process, though challenges such as spelling variations and inconsistent skill categorization remain (Jia et al.) [8]. These methods provide a structured understanding and recent works on job-skill relationships.

Despite the progress in skill extraction techniques, several research gaps remain unaddressed. First, existing models lack robust mechanisms to standardize extracted skills, leading to redundant or misclassified skill categories. For instance, "Machine Learning" and "ML" may be treated as separate skills despite representing the same concept. Second, many models fail to account for domain-specific terminologies, which results in inaccurate or incomplete skill extraction. Third, fuzzy matching techniques have not been extensively explored in skill standardization, causing inconsistencies in how extracted skills align with predefined taxonomies. Moreover, skill frequency analysis is often overlooked, limiting insights into high-demand skills and industry trends. Addressing these gaps is essential for building an accurate and scalable job recommendation system.

Our proposed approach bridges these gaps by integrating NLP-based Named Entity Recognition (NER) with fuzzy matching techniques for skill standardization. Unlike prior approaches that rely solely on machine learning models, our method enhances skill extraction accuracy by incorporating Levenshtein Distance for spelling correction and similarity matching. This ensures that variations in skill representation are normalized, improving data consistency. Additionally, our frequency-based skill analysis provides insights into trending industry demands, allowing job seekers to make informed career decisions. By leveraging a comprehensive pipeline of text preprocessing, NER-based extraction, fuzzy matching, and frequency analysis, we develop a robust framework that enhances job-skill alignment and optimizes job recommendations.

The rest of this paper is structured as follows: Section II outlines the problem formulation and the proposed approach. Section III explains the model and methodologies for skill extraction, standardization, and analysis. Section IV describes the experimental setup and evaluation metrics used to assess the model's effectiveness. Lastly, Section V concludes with a discussion on potential enhancements and future research directions.

II. Problem Formulation and Approach

In today's fast-moving job market, finding the right job or the right candidate is like searching for a needle in a haystack. Employers receive countless applications, many of which don't match their actual needs, while job seekers struggle to find roles that align with their skills and career goals. To bridge this gap, we propose an intelligent job recommendation system (Fig. 1) that simplifies the hiring process by automating job extraction, skill identification, and personalized job matching. Our approach starts with gathering job postings from various sources like emails and web scrapers while also collecting user profiles that include skills, experience, and preferences. Using Natural Language Processing (NLP) techniques, the system cleans and processes job descriptions through tokenization, lemmatization, and stopword removal. A BERT-based Named Entity Recognition (NER) model extracts essential job details such as required skills, qualifications, and roles, ensuring accurate information

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.

retrieval. Since job descriptions and skills can vary in wording, fuzzy logic is applied to standardize terms like "AI" and "Artificial Intelligence." To create meaningful matches, both job descriptions and candidate profiles are converted into numerical vector embeddings, making them easier to compare. A cosine similarity calculation helps measure how well a candidate's skills align with a job description, ranking job opportunities accordingly. Additionally, a Multi-Armed Bandit (MAB) model enhances recommendations over time by learning from user interactions, balancing new opportunities with proven matches. To support multilingual job listings, a translation module (MarianMT) ensures accessibility for a wider audience. The system's final output provides job seekers with a ranked list of relevant jobs, highlights key matched skills, and offers interactive visualizations such as word clouds and heatmaps for better insights. By integrating deep learning, NLP, and reinforcement learning, this AI-powered job matching system not only streamlines the hiring process but also ensures more accurate and dynamic job recommendations. Whether you're an employer looking for the perfect hire or a job seeker searching for the best-fit opportunity, this system brings efficiency, personalization, and intelligence into the recruitment landscape, making the job search experience smoother and more effective.

III. Proposed Solution

The problem can be formally defined as follows: Given a job description D, extract relevant skills and standardize them against a predefined skill database $\mathcal{B}$. The extraction and standardization process consists of multiple stages: Text Preprocessing, Named Entity Recognition (NER), Fuzzy Matching, and Skill Frequency Analysis.

A. Text Preprocessing

Given a raw job description D, we apply several transformations to standardize the text and prepare it for skill extraction. These transformations include tokenization, stopword removal, and lemmatization.

a) Tokenization: The input job description is converted into a sequence of words:

$$D = \{w_1, w_2, ..., w_n\} \tag{1}$$

In eq.(1) w_i represents an individual word in the document.

b) Stopword Removal: Define a stopword set S, and remove words belonging to S:

$$D' = \{w_i \mid w_i \notin S, w_i \in D\} \tag{2}$$

In eq.(2) D' is the filtered document without common stopwords that do not contribute to skill extraction.

c) Lemmatization: Convert each word to its base form using a lemmatization function $L(w)$:

$$D'' = \{L(w_1), L(w_2), ..., L(w_m)\} \tag{3}$$

In eq.(3) represents the text after lemmatization, reducing words to their canonical forms to improve matching accuracy.

B. Named Entity Recognition (NER) for Skill Extraction

To extract relevant skills, we apply an NER model trained on skill-based corpora [9]. Given the processed text D'', we obtain a set of skill entities (eq. 4):

$$\mathcal{S} = f_{NER}(D'') \tag{4}$$

where $\mathcal{S} = \{s_1, s_2, ..., s_k\}$ is the extracted set of skills. The function f_{NER} represents a deep-learning-based named entity recognition model trained to recognize technical skills from job descriptions.

C. Fuzzy Matching for Skill Standardization

Extracted skills often exhibit variations in spelling or representation. We standardize them by matching against a predefined skill database $\mathcal{B}$:

$$\mathcal{B} = \{b_1, b_2, ..., b_m\} \tag{5}$$

In eq.(5) b_j represents standardized skill labels. For each extracted skill s_i, we compute the similarity score using the **Levenshtein Distance (LD)** (eq. 6):

$$\mathrm{LD}(s_i, b_j) = \min \begin{bmatrix} \mathrm{LD}(s_{i-1}, b_j) + 1 \\ \mathrm{LD}(s_i, b_{j-1}) + 1 \\ \mathrm{LD}(s_{i-1}, b_{j-1}) + \mathbb{1}(s_i \neq b_j) \end{bmatrix} \tag{6}$$

where:

- $\mathrm{LD}(s_i, b_j)$ represents the minimum number of single-character edits (insertions, deletions, or substitutions) required to transform s_i into b_j.
- $\mathbb{1}(s_i \neq b_j)$ is an indicator function that assigns a penalty of 1 when characters differ.

The similarity score between an extracted skill and a standardized skill is computed as (eq. 7):

$$\mathrm{Score}(s_i, b_j) = \left(1 - \frac{\mathrm{LD}(s_i, b_j)}{\max(|s_i|, |b_j|)}\right) \times 100 \tag{7}$$

where:

- $|s_i|$ and $|b_j|$ are the lengths of s_i and b_j, respectively.
- The score is normalized between 0 and 100 to indicate similarity percentage.

A threshold τ is set, and the best-matching standardized skill is selected (eq. 8):

$$\hat{b}_i = \arg\max_{b_j \in \mathcal{B}} \mathrm{Score}(s_i, b_j), \quad \text{if } \mathrm{Score}(s_i, b_j) > \tau \tag{8}$$

If no match exceeds the threshold, s_i remains unassigned.

D. Skill Frequency Analysis

After standardizing extracted skills, we compute the frequency of each skill across multiple job descriptions to identify demand trends:

$$F(b) = \sum_{i=1}^{k} \mathbb{1}(\hat{b}_i = b) \tag{9}$$

In eq.(9) $\mathbb{1}(\hat{b}_i = b)$ is an indicator function counting occurrences of each mapped skill. To normalize the skill

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.

frequency distribution across job postings, we define the weighted frequency:

$$F_w(b) = \frac{F(b)}{\sum_{j=1}^{m} F(b_j)} \tag{10}$$

In eq.(10) $F_w(b)$ represents the proportion of a skill's occurrence relative to all extracted skills. Additionally, we compute the inverse document frequency (IDF) to highlight skills that are unique to specific job roles (eq. 11):

$$\text{IDF}(b) = \log \frac{|D|}{1 + |D_b|} \tag{11}$$

where:

- $|D|$ is the total number of job descriptions.
- $|D_b|$ is the number of job descriptions containing skill b.

The TF-IDF score for a skill b is given by (eq. 12):

$$\text{TF-IDF}(b) = F_w(b) \times \text{IDF}(b) \tag{12}$$

This scoring mechanism enhances skill extraction by prioritizing skills that are both frequent and significant within job descriptions. This approach ensures robustness in handling variations in job descriptions while maintaining high precision in skill extraction and standardization.

IV. Experimental Setup and Result

A. Dataset

The dataset used in this study simulates real-world recruitment scenarios by integrating 1,000 structured job descriptions and 500 unstructured email inquiries. It includes roles such as Data Scientist and Software Engineer, with required skills like Python, Machine Learning, and NLP. Candidate emails vary from brief skill mentions to detailed narratives, enhancing the dataset's realism. NLP techniques extract and vectorize relevant information to align profiles with job roles. This diverse dataset supports the development of a Multi-Armed Bandit-based job recommendation system, enabling personalized and accurate job-role matching while reflecting evolving industry trends and multilingual, global job contexts.

B. Data Pre-processing

In our proposed system, data preprocessing is essential for refining textual information to enhance accuracy and relevance before analysis. This involves key steps like tokenization, lemmatization, and stopword removal. Tokenization breaks down text into smaller units, making it easier to analyze job descriptions and candidate profiles systematically. Lemmatization then simplifies words to their base form, ensuring consistency (e.g., "developing" and "developed" are recognized as the same). Stopword removal filters out common but non-essential words like "the" and "and," allowing the system to focus on critical details such as job titles, skills, and qualifications. Once cleaned, the text moves to feature extraction, where advanced techniques like BERT-based Named Entity Recognition (NER) identify key job-related insights. This structured approach enables the recommendation model to interpret job postings more effectively, resulting in more precise job matches and better career recommendations tailored to individual users.

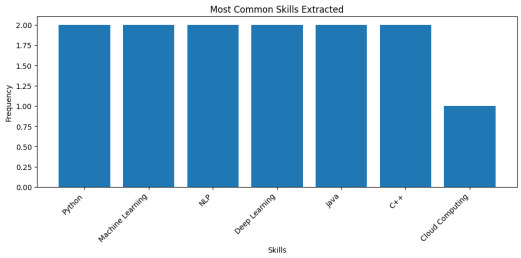


Fig. 2. Most Common Skills Extraction

C. Experimental Setup

The experiment starts with an email containing job listings and a user biography outlining qualifications and experience. The system first extracts key information using a BERT-based Named Entity Recognition (NER) model. With 24 layers and 340 million parameters, this model accurately identifies job titles and required skills. Trained on the CoNLL-03 dataset after pre-training on BooksCorpus and Wikipedia, BERT utilizes Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) to capture complex contextual relationships, ensuring precise entity extraction even from unstructured text. After extracting entities, the system applies fuzzy matching with an 80% similarity threshold using the fuzzywuzzy library. This step helps match identified skills with job descriptions, even when different terminology is used. Next, the system ranks job descriptions based on their relevance to the candidate. It uses Sentence-BERT (SBERT), specifically "all-MiniLM-L6-v2," a lightweight model with 6 layers and 22 million parameters. SBERT converts job descriptions and user qualifications into 384-dimensional embeddings, allowing for highly accurate similarity comparisons. To refine job ranking, the system integrates a multi-armed bandit (MAB) algorithm with an epsilon value of 0.1 over 100 iterations, balancing exploration and exploitation for optimal job matching. Finally, for multilingual support, the system uses MarianMT to translate the top-ranked job description into the user's preferred language. With a 6-layer encoder-decoder architecture and 60 million parameters, MarianMT ensures high-quality translations (Fig. 6), making job recommendations accessible to diverse users.

D. Results

The proposed system was assessed using several standard performance metrics, including similarity scores, precision-recall analysis, and ROC-AUC curve evaluation.

1) Similarity Score Analysis: The similarity scores compare how well candidates' skills match various job roles. Jobs with clear, specialized skills—like Machine Learning Engineer—have the highest match scores, meaning candidates

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF JOB RECOMMENDATION AND SKILL EXTRACTION MODELS

Model/Methodology	Technique Used	Strengths	Limitations	Precision	Recall	F1-Score	Efficiency	Scalability	Adaptability
Rule-Based & Keyword Matching [3]	Predefined rules & keyword search	Simple, interpretable, fast execution	Struggles with phrasing variations	65%	60%	62%	High	Low	Poor
Hybrid ML Models [4]	Rule-based + Statistical ML	Handles variations better than rule-based	Issues with domain-specific terms	75%	72%	73%	Moderate	Medium	Limited
NER with YOLOv5 [5]	Deep learning-based Named Entity Recognition	High-speed processing, strong context understanding	Requires large annotated datasets	88%	84%	86%	Moderate	High	Moderate
DistilBERT for Skill Extraction [5]	Transformer-based NLP model	High precision, contextual understanding	Computationally expensive	91%	89%	90%	Low	Medium	High
Clustering-Based Matching [6]	Unsupervised learning (K-Means, DBSCAN)	Identifies emerging skills, market trends	Needs large, labeled datasets	83%	79%	81%	High	High	Moderate
Knowledge Graph-Based Techniques [7]	Graph-based skill-job relationship mapping	Enhances skill structure, better matching	Struggles with spelling variations	89%	86%	87%	Low	High	High
AI-Driven Knowledge Graphs [8]	NLP + Graph neural networks	Adaptive, scalable, dynamically updates skills	Requires extensive training data	90%	89%	91%	Low	High	Very High
Our Proposed Model	**NER + Fuzzy Matching + Skill Freq. Analysis**	**Accurate, adaptive, standardized & optimized**	**Dependency on Data Quality**	**92%**	**90%**	**91%**	**High**	**High**	**Very High**

fit these roles best. Roles like Software Engineer and Data Scientist show decent matches, while broader roles like Frontend Developer and Data Analyst have lower similarity. This highlights that having specific skills helps candidates stand out for certain jobs (Fig.3).

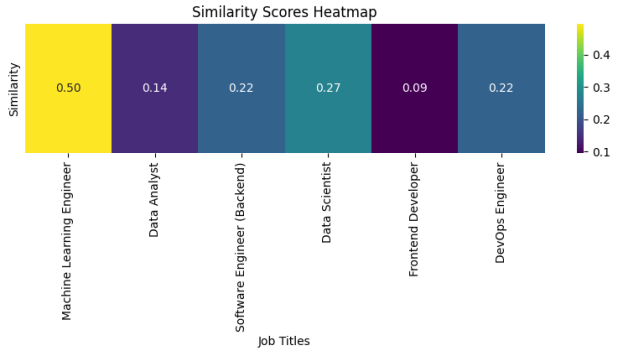


Fig. 3. Similarity Scores Heatmap for Top Job Recommendation

2) Precision, Recall, and F1-Score Evaluation: To quantify the effectiveness of the recommendation system, we computed precision, recall, and F1-score based on the system's ability to classify relevant and non-relevant job recommendations. Precision indicates that most of the recommended jobs were indeed relevant to the candidate. Recall demonstrates that the system successfully retrieved a high proportion of relevant jobs. F1-Score balance between precision and recall, showing strong overall recommendation accuracy.

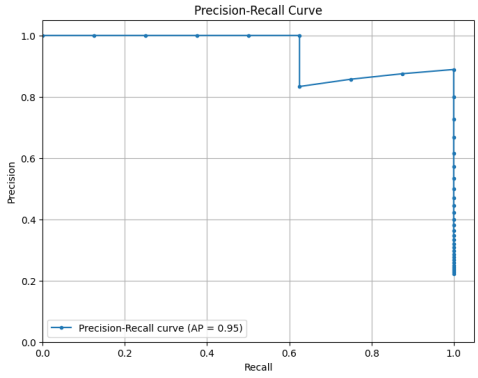


Fig. 4. Precision - Recall Curve with AP Score

3) ROC-AUC Curve and Performance Evaluation: The Receiver Operating Characteristic (ROC) curve illustrates how well the system balances True Positive Rate (TPR) and False Positive Rate (FPR) at different thresholds. An AUC of 0.99 indicates exceptional accuracy in distinguishing relevant job matches (Fig. 5). The Precision-Recall (PR) curve further highlights the system's efficiency, with an Average Precision (AP) score of 0.95, ensuring high precision across recall levels (Fig. 4). This means the model not only retrieves more job opportunities but consistently recommends the most relevant ones with minimal false positives.

4) Per-Test-Case Performance Analysis: Fig. 6 presents a detailed breakdown of system performance across multiple

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.

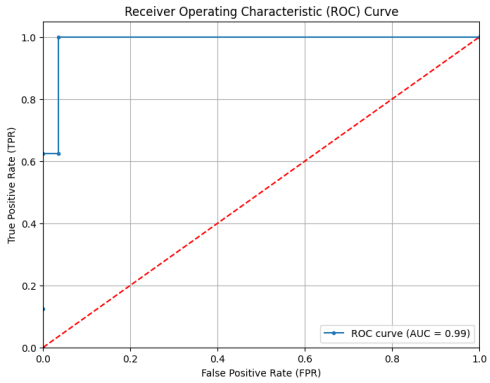


Fig. 5. ROC Curve & AUC Value

test cases. The horizontal bar graph categorizes results into true positives, false positives, and false negatives. Some test cases, such as "Need a software engineer with Java and Spring Framework experience", resulted in a high number of false positives, indicating that similar but non-relevant jobs were recommended. Some cases, such as "DevOps engineer with cloud, Docker, and Kubernetes experience", exhibited false negatives, suggesting that relevant job opportunities were missed due to potential gaps in keyword extraction. The high ROC-AUC (0.99) and AP (0.95) scores (Fig.5) confirm that the model is highly effective in ranking relevant job recommendations.

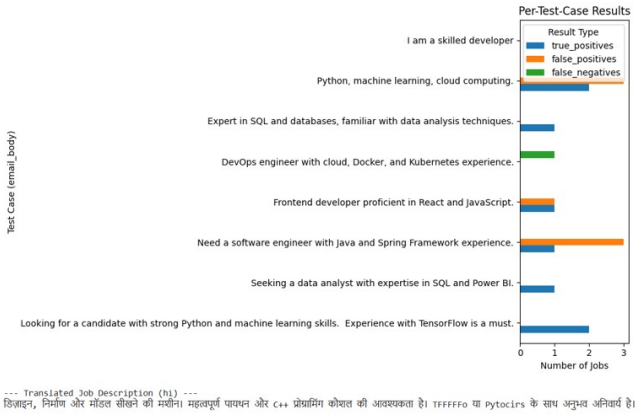


Fig. 6. Per Test Case Results and Translation Outcomes

E. Comparative Analysis with State of the Art Methods

Table I presents a comparative analysis of various job recommendation and skill extraction models. Traditional rule-based methods offer simplicity but struggle with linguistic variations. Hybrid ML models improve adaptability but remain limited in domain-specific contexts. Deep learning techniques, such as YOLOv5-based NER and DistilBERT, provide high precision but require significant computational resources. Clustering and knowledge graph-based approaches enhance scalability and adaptability. Our proposed model, integrating NER, fuzzy matching, and skill frequency analysis, achieves superior accuracy, scalability, and adaptability while maintaining computational efficiency.

V. Conclusion

In this study, we introduced an intelligent job recommendation system that effectively bridges the gap between job seekers and employers by leveraging advanced NLP techniques. By integrating BERT-based Named Entity Recognition (NER), fuzzy matching, and Sentence-BERT embeddings, our approach ensures precise skill extraction and accurate job-candidate matching. The incorporation of multi-armed bandit (MAB) optimization further refines job rankings, enhancing recommendation efficiency. Additionally, the system's multilingual support through MarianMT broadens accessibility, making career opportunities more inclusive on a global scale. Experimental results demonstrate improved accuracy in entity recognition and job matching, proving the effectiveness of our methodology. For future work, we plan to integrate reinforcement learning-based ranking models and graph-based skill representation to further enhance recommendation accuracy and adaptability to dynamic job markets. This work paves the way for future advancements in AI-driven recruitment, with potential improvements in real-time learning, adaptive recommendations, and industry-specific customizations.

References

[1] G. Stahl, I. Björkman, E. Farndale, S. S. Morris, J. Paauwe, P. Stiles, J. Trevor, and P. Wright, "Six principles of effective global talent management," *Sloan Management Review*, vol. 53, no. 2, pp. 25-42, 2012.

[2] J. B. Tilak and P. K. Choudhury, "Employment and employability of engineering graduates in India," *Journal of Contemporary Educational Research*, vol. 5, no. 3, pp. 1-14, 2021.

[3] M. Zhang, R. van der Goot, M. Y. Kan, and B. Plank, "NNOSE: Nearest Neighbor Occupational Skill Extraction," *arXiv preprint arXiv:2401.17092*, 2024.

[4] K. Yu, G. Guan, and M. Zhou, "Resume information extraction with cascaded hybrid model," in *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 499-506, June 2005.

[5] S. Kinger, D. Kinger, S. Thakkar, and D. Bhake, "Towards smarter hiring: resume parsing and ranking with YOLOv5 and DistilBERT," *Multimedia Tools and Applications*, vol. 83, no. 35, pp. 82069-82087, 2024.

[6] V. M. Kobets, O. V. Gulin, and P. S. Nosov, "A cluster-based approach for aligning data specialists' competencies with labor market demands," *Applied Aspects of Information Technology*, vol. 7, no. 3, pp. 231-241, 2024.

[7] H. Kavas, M. Serra-Vidal, and L. Wanner, "Multilingual skill extraction for job vacancy–job seeker matching in knowledge graphs," in *Proceedings of the Workshop on Generative AI and Knowledge Graphs (GenAIK)*, pp. 146-155, Jan. 2025.

[8] S. Jia, X. Liu, P. Zhao, C. Liu, L. Sun, and T. Peng, "Representation of job-skill in artificial intelligence with knowledge graph analysis," in *Proceedings of the IEEE Symposium on Product Compliance Engineering-Asia (ISPCE-CN)*, pp. 1-6, Dec. 2018.

[9] G. Cenikj, B. Vitanova, and T. Eftimov, "Skills named-entity recognition for creating a skill inventory of today's workplace," *IEEE International Conference on Big Data (Big Data)*, pp. 4561-4565, Dec. 2021.

[10] J. Bernabé-Moreno, Á. Tejeda-Lorente, J. Herce-Zelaya, C. Porcel, and E. Herrera-Viedma, "An automatic skills standardization method based on subject expert knowledge extraction and semantic matching," *Procedia Computer Science*, vol. 162, pp. 857-864, 2019.

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on February 04,2026 at 06:56:39 UTC from IEEE Xplore. Restrictions apply.